

Avatar Interpreter: Improving Classroom Experiences for Deaf and Hard-of-Hearing People Based on Augmented Reality

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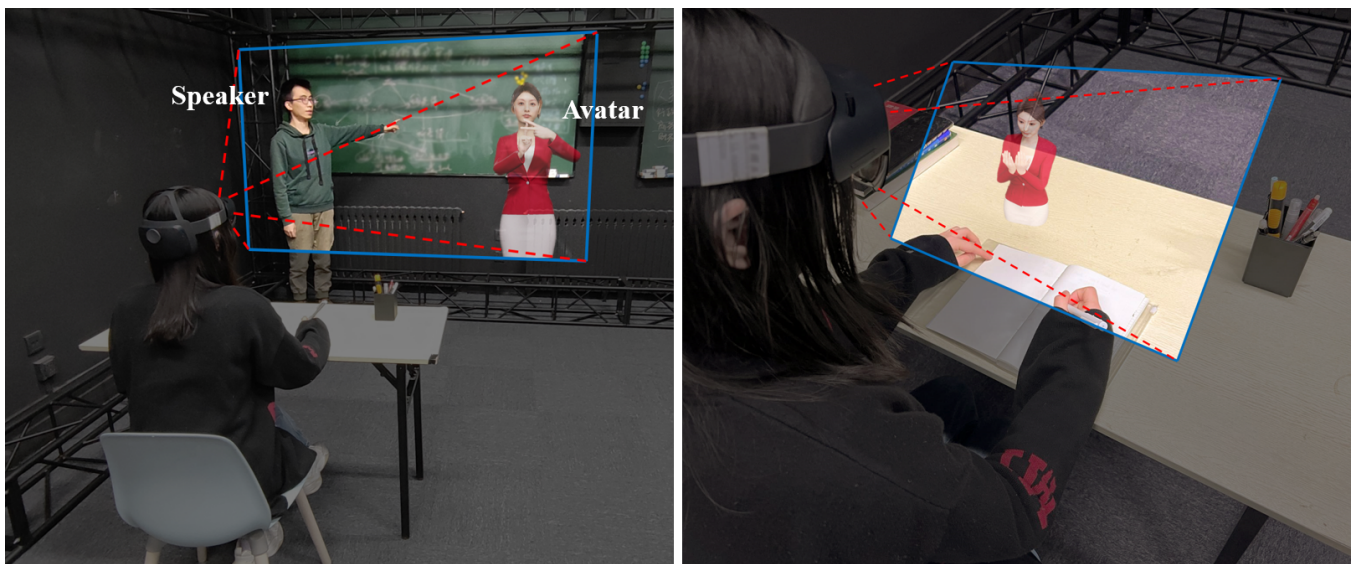


Figure 1: Demonstration of Avatar Interpreter: Users wear a HoloLens 2 and watch the blackboard (left) and the book (right) while Avatar Interpreter provides the user with real-time sign language interpretation from the speaker. (The user and the speaker are in real environment, the avatar is shown in HoloLens2)

ABSTRACT

Deaf and hard-of-hearing (DHH) people experience difficulties in accessing education. One reason is that they miss out on oral information in the classroom. Although some classrooms are equipped with sign language interpreters specifically for deaf students, DHH students have trouble switching their gaze and attention when they cannot see the interpreter and the visual information in the classroom simultaneously. To address this challenge, this paper

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develops Avatar Interpreter, a visual interface for synchronized sign language interpreters on an augmented reality head-mounted display. Avatar Interpreter can be displayed at different sizes and in different locations in space to fit different scenarios, helping deaf students to better receive sign language information and enhancing the classroom experience for deaf students. In the work presented in this paper, we discuss our prototype, make design recommendations, and discuss configuration and implementation. Finally, we propose questions and research methods for an upcoming user study.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility design and evaluation methods; Mixed / augmented reality.**

KEYWORDS

Accessibility, DHH, AR, Avatar, Sign Language

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1 INTRODUCTION & RELATED WORK

DHH people have difficulty receiving audio information in the same way as hearing people, which makes it difficult for them to access information in the classroom. DHH people must switch back and forth between sign language interpreters or captions and other visual information sources (slides, chalkboard, notes, lecturer, etc.), which means that they miss or lose some information. Helping DHH people reduce this visual distraction can enhance their classroom experience. Kushalnagar et al. [10] proposed a speech-to-text scheme with real-time tracking that follows the position of the classroom presenter and displays captions more intelligently to help DHH people have a better classroom experience. Gugenheimer et al. [5] projected the sign language interpreter's gestures onto the upper body of the speaker so that the DHH person could understand what the speaker was expressing while looking at the speaker.

With the increasing popularity and reliability of smart glasses, some accessibility research has focused on using smart glasses to help DHH individuals understand speech information and enhance the experience of DHH individuals. Kercher et al. [8] used an augmented reality head-mounted system to keep the sign language interpreter within the wearer's field of view, thereby improving the experience of visiting a planetarium for deaf people. Peng et al. [15] developed an augmented reality head-mounted display (HMD)-based "speech bubble" prototype, which translates the speaker's speech into text and superimposes it next to the speaker's head in the form of a text bubble. Jain et al. [7] investigated captioning design for HMDs to enable DHH users to achieve communication access even while on the move. Miller et al. [13] explored the potential of using smart glasses in the classroom to help deaf college students participate in the lessons of hearing students by displaying captions or sign language videos on smart glasses. Vinayagamoorthy et al. [18] used augmented reality HMDs to improve the experience of watching TV for DHH people and studied full-body and half-body sign language interpreters versus the traditional sign language interpreter display format. However, the interpreter is only a plane in space and does not truly have a three-dimensional effect.

Previous research [11, 13, 17] has demonstrated that DHH individuals have limited reading and comprehension skills for captioning, and for them, using sign language to convey information is more friendly than captioning, especially in a classroom setting. References [2, 3] investigated the use of an avatar as a sign language interpreter for speech or text-to-sign language conversion. Based on these works, we designed "Avatar Interpreter", an augmented reality-based interface for synchronized sign language visualization. In this paper, we describe our process of designing this prototype

based on the augmented reality head-mounted display HoloLens 2 [16], and we describe how our prototype can be applied in the classroom to help DHH individuals reduce the visual distraction caused by switching back and forth between sign language and other visual information to improve the classroom experience for DHH individuals. We also explore the design of different display methods, as well as adapting the display format, position, and size to different situations. Finally, we present questions and research proposals for an upcoming user study.

2 CONCEPT

The idea of this paper is to build a humanoid avatar and combine the advantages of augmented reality technology to have the avatar displayed in the correct location to provide synchronized sign language translation for DHH users to help them understand speech information. The visual interface first needs to provide DHH users with accurate speech-to-sign language translation that can be clearly presented to the user. Second, the sign language information can be continuously displayed in the user's field of view when the user switches between sources of visual information without requiring the user to turn his or her head to find it. For example, in a classroom, when the user is watching a presentation, the lecturer's voice message will be translated into a sign language message that is displayed alongside it and can be easily seen by the user; when the user looks down to take notes, the sign language message should also move and continue to be displayed in the user's field of view. Finally, the content displayed in our visual interface should be in the right place, not blocking the user's view or the user's current focus. Additionally, the depth of the displayed information should be adapted to the surrounding environment as well as possible to ensure that the information can be observed naturally by the user.

3 DESIGN & IMPLEMENTATION

3.1 Early User Research

To better understand the issues DHH students face in the classroom, we conducted a preliminary user study through structured interviews and a needs survey. Three individuals participated in our initial user study: one was a teacher at a school for the deaf who had normal hearing, and the other two were DHH high school students with severe hearing loss; all three could use Chinese sign language. We first introduced participants to AR technology and Microsoft HoloLens 2 and demonstrated some HoloLens 2-based applications. Next, participants described to us current classroom scenarios for DHH students and presented some current difficulties they encountered and how they coped with them. Then, we discussed with participants how augmented reality technology and AR HMDs could be used to improve their experience; and finally, we discussed with participants their ideal visualization interface.

3.2 Design

Avatar appearance: Previous works [1, 9, 14] have shown that sign language avatars are affected by the Uncanny Valley. When the sign language avatars are not sufficiently similar to real humans, DHH users are more concerned with the naturalness of the facial expressions, appearance, and movements of the sign language avatars than the accuracy of the sign language. Therefore, the avatars need



Figure 2: Flexible Mode: in the left column, the user looks at the blackboard, and in the right column, the user looks at the book, the top row is the external view, and the bottom row is the view in Hololens 2.

to have a high sense of realism as well as diverse and natural facial expressions so that DHH users can feel that they are more natural and closer to real sign language interpreters.

Sign language actions: Sign language movements generally consist of hand patterns (including finger movements), body positions and movements, which are combined according to certain rules to describe information. Therefore, the accuracy of avatar movements affects the accuracy of the sign language, and the transitions between different movements should be continuous and smooth. Additionally, the inclusion of facial expressions and mouth shapes in sign language can improve the understandability and perceived quality of the animation and can make the avatar look more natural [6].

Display modes: We designed “Avatar Interpreter” to ensure that users can easily see the avatar while focusing on other visual information, thus enhancing the DHH user experience in the classroom. We designed two display modes to present the sign language information of the avatar to users.

1. **Flexible Mode:** In this mode (Figure 2.), we set the position of the avatar for different gaze behaviors of the user in the classroom. When the user is looking at the lectern, the avatar interpreter is the same size as a normal person. In this scenario, the avatar interpreter should not block the speaker, nor should it block the content display area (TV, projection area or blackboard). Therefore, we place the avatar interpreter next to the speaker or near the content display area, similar to the situation in which a live interpreter is used. When the user is looking down at the desktop, we shrink the avatar and place it on the desktop. In this case, the avatar interpreter is in the user’s field of view and does not block their main area of focus (e.g., a book or laptop screen).

2. **Fixed Mode:** Referring to previous studies on the human visual system and peripheral vision [12], the center of the visual field below 2° eccentricity is called the foveal vision area, which has the highest visual acuity, and the visual field of the fovea and periphery (approximately 8° radius) is called the central visual field. Anything out of the foveal visual field (2° eccentricity) is called peripheral vision. In fixed mode (Figure 3.), we fix the avatar at an angle from the center of the user’s field of view and always follow the user’s

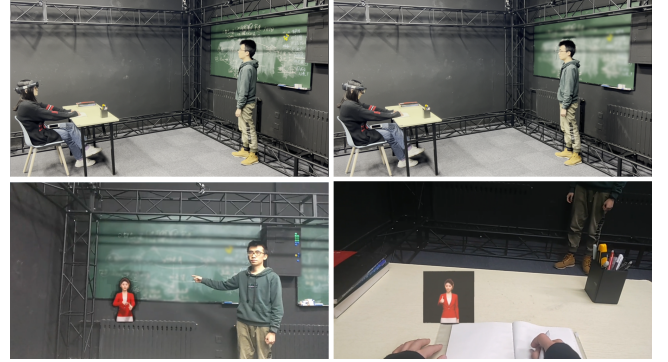


Figure 3: Fixed Mode: in the left column, the user looks at the blackboard, and in the right column, the user looks at the book, the top row is the external view, and the bottom row is the view in Hololens 2.

head movement. This allows the user to use the central field of view to perceive highly important visual information (e.g., presentation, books) while using peripheral vision to access the sign language information provided by the avatar. This design is similar to traditional TV programs, where the sign language interpreter is usually displayed in a small window mode in the bottom left or right corner of the screen.

Implementation: We reconstructed the face with high accuracy using a multi-illumination light field generation and multiview acquisition system for the material information and a geometric model of the face. We also captured 40 different blendshape models for the face-driven blendshape model as the base expression. We used Unreal Engine version 4.26 to develop our visualization interface. The entire system runs on a PC (Intel Core i9-7900X, NVIDIA GeForce RTX3080, 32 GB RAM and 1 TB SSD) and is projected to Microsoft HoloLens 2 ($43^\circ \times 29^\circ$ field of view, resolution 1440×936). We used OptiTrack and data gloves for the motion capture of sign language movements and an iPhoneXR with Live Link Face [4] installed to drive the avatar’s facial expressions.

4 METHOD

Our research process consists of two parts: the first part aims to refine the study of the two display modes; the second part aims to investigate the usability of our visualization interface and user preferences for the two display modes.

In the first part, we plan to start with a questionnaire to collect participants’ demographic information and background and to understand their degree of hearing loss and experience with AR HMD. Then, we will clarify the classroom scenario and present our prototype system to the participants, asking them to experience the two different display modes with various predefined conditions; we will ask the participants to compare the display scenarios and ask them about their preferences. The conditions we plan to compare in the two modes are described below.

Flexible Mode:

- The position of the avatar interpreter when the line of sight is on the podium.

- The size, position, and display form (full body or half body) of the avatar interpreter when the line of sight is on the table.

Fixed Mode:

- The display form of the avatar interpreter: floating in 3D or displayed as a flat surface in 2D (similar to the small window showing the interpreter in a TV show).
- The display position, size and transparency of the avatar interpreter.
- The background of the avatar interpreter: whether a background color is needed to make the avatar stand out and reduce distraction from the surroundings.

In the second part, we will simulate a classroom situation by asking participants to complete some tasks we designed to investigate the usability of our prototype and to compare the ergonomics of the flexible avatar model and the fixed model and participants' preferences for them. We will include a condition in which there is a life-size avatar interpreter at the podium, similar to the real classroom situation, as a control. Thus, we plan to compare the following three conditions:

1. Avatar interpreters on the podium only
2. Flexible mode
3. Fixed mode;

5 DISCUSSION

During the early user research, all three participants expressed interest in AR technology, and they all agreed that AR technology showed potential in terms of being useful for improving the classroom experience for DHH people. In terms of avatar images, one participant suggested that images such as cartoons or animals could be used, but he also agreed that high-fidelity images could more accurately convey expressions, which could help DHH individuals understand the information more accurately. In terms of display, our design covered most of the ideas proposed by the participants, was sufficiently diverse, and kept some cases with disagreement (e.g., whether the avatar is shown with a half or full body on the desktop) as different conditions to be compared in the upcoming user study.

Expansion: We also discussed the extension of Avatar Interpreter so that it can be applied to more situations; e.g., in the multi-person group session mode, the avatar can appear in the position of different speakers and indicate the speaker's position while providing sign language interpretation. When the speaker is far away from the user (e.g., in a large lecture), sign language interpretation can be continuously provided to the user by zooming in on the avatar or using the fixed mode.

Limitations: There is a lack of available Chinese sign language (CSL) real-time translation database support, so we cannot validate the performance of Avatar Interpreter in real-world applications, and we can only drive Avatar Interpreter through motion capture and facial capture to study the usability of our display interface in the form of actor performances, advance recordings, etc.

We used HoloLens 2 as our AR HMD, but it has a limited field of view and resolution, as well as color separation issues that may affect the presentation of Avatar Interpreter.

6 CONCLUSION & FUTURE WORK

We present Avatar Interpreter, a synchronous sign language translation visualization interface for AR HMDs to help improve the experience of DHH users in the classroom. In this paper, we describe the concept and design ideas of the interface and implement a prototype system on Microsoft HoloLens 2. Then, we present the methodology and plan for an upcoming user study. In future work, we plan to conduct two user studies to further refine the design of both display modes. At the same time, we plan to evaluate the usability of our system and the impact of the two display modes on the user experience. We hope to extend Avatar Interpreter to other application scenarios.

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